

Fluid: towards transparent, self-explanatory research outputs

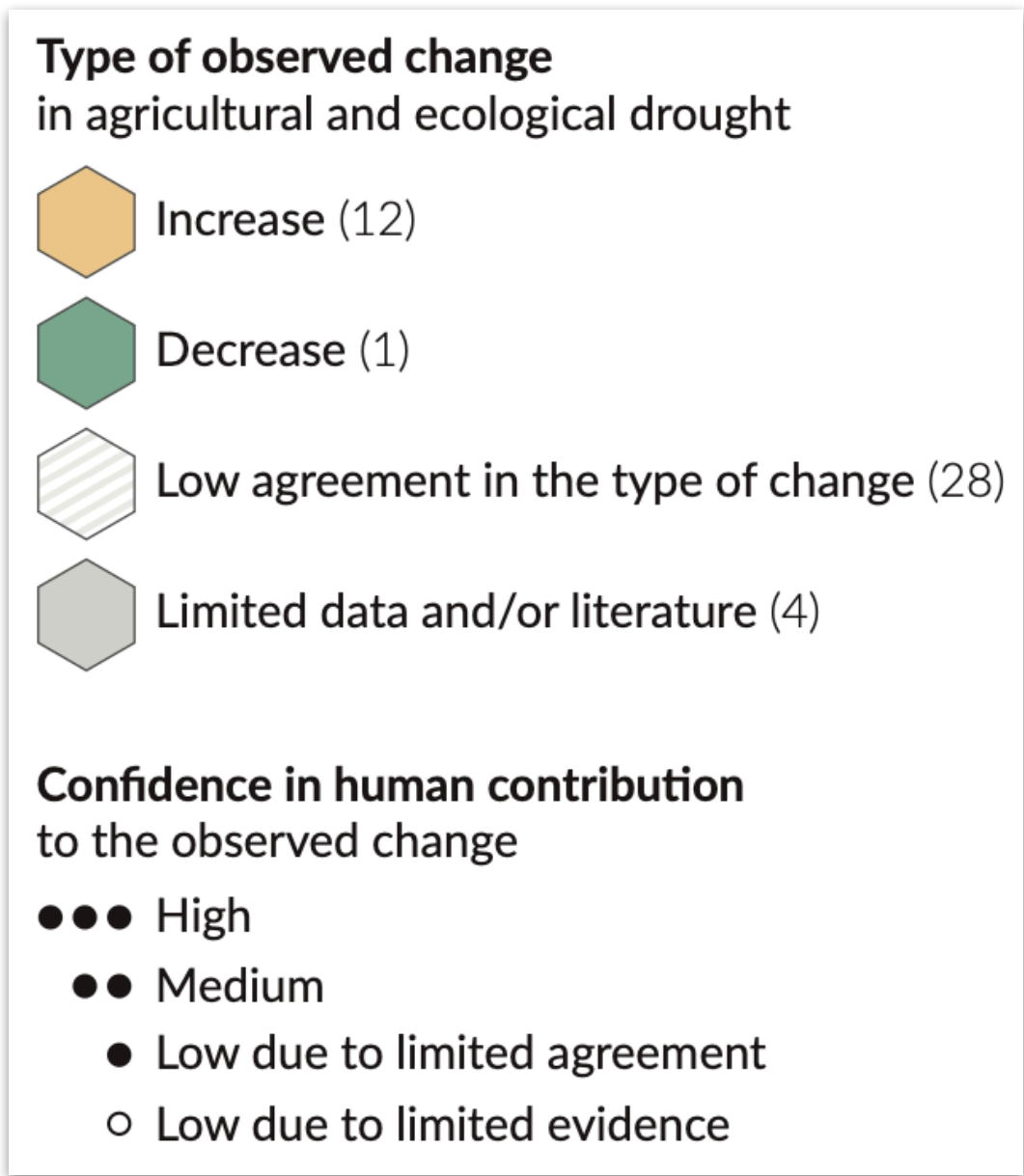
Joe Bond^{*}, Cristina David^{*}, Minh Nguyen^{*}, Roly Perera^{*†}

^{*}University of Bristol

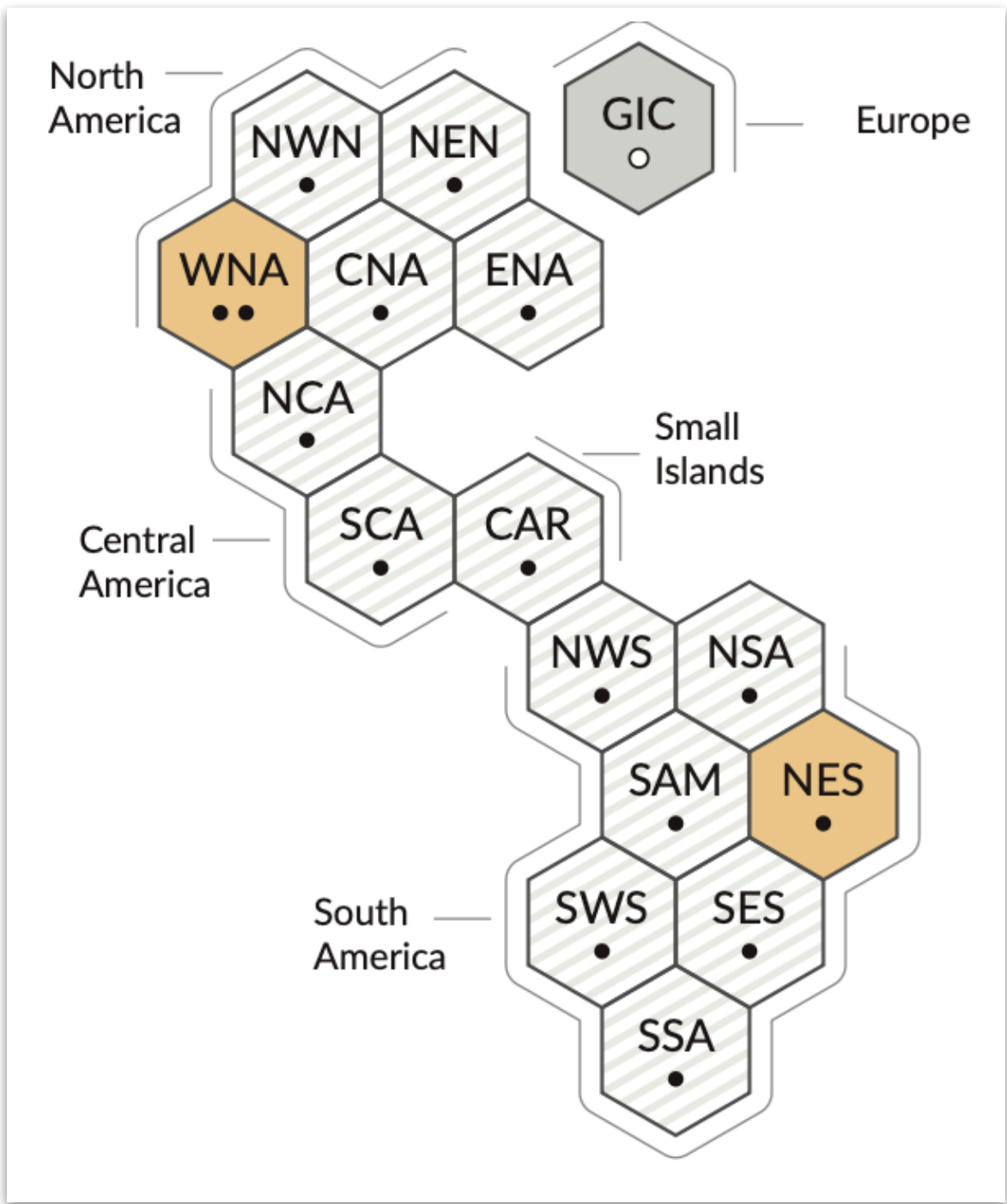
[†]Institute of Computing for Climate Science, University of Cambridge

Understanding, trusting and evaluating visual information

We form beliefs and make decisions using data, curated into charts, figures and text



Legends/captions of limited value



Informative in a certain sense but

- what are we looking at?
- where did it come from?
- why should we should trust it?

Challenge for policy makers, scientists, journalists, educators (and everyone else)

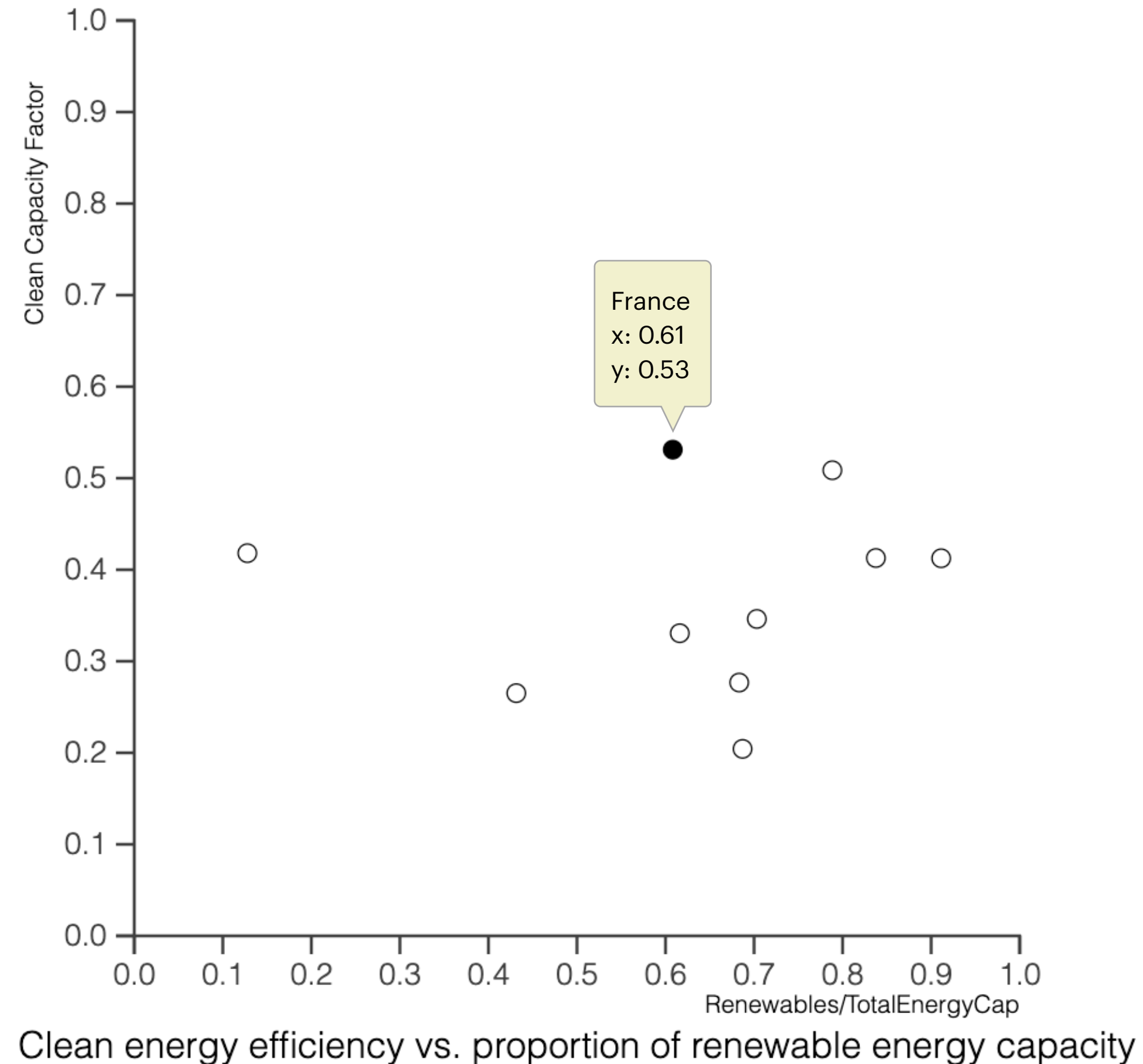
Familiar problem for traditional print media, but what about digital media?

Can PL help?

Heating of the climate system has caused global mean sea level rise through ice loss on land from ocean warming. Thermal expansion explained 50% of sea level rise during 1971–2018, contributed 22%, ice sheets 20% and changes in land-water storage 8%. The rate of ice-sheet loss was four times faster between 1992–1999 and 2010–2019. Together, ice-sheet and glacier mass loss were the

Supporting text mentions computed data

Towards transparent, self-explanatory data visualisation



Questions we might ask

What year is this data from?

Is each point a country?

What do x and y represent?

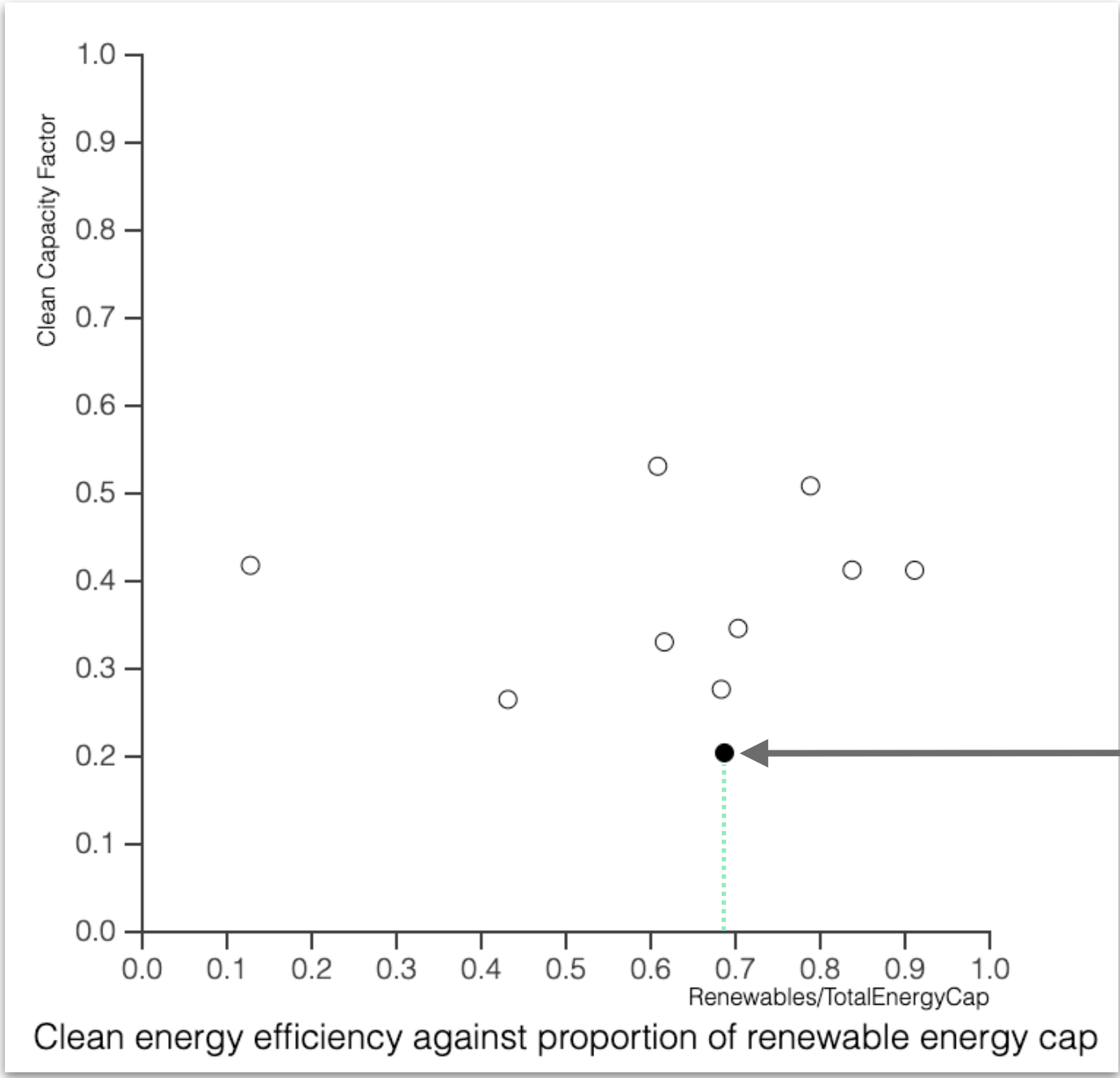
What energy types count as “renewable”?

Viz libraries provide tooltips which clarify the **outputs**, but what about the relationship to **inputs**?

Data and source code help but reverse-engineering relationships is not easy

Demo: related inputs

Related inputs operator $\Delta \nabla$
Not idempotent



Mediating output answers that question and induces a demand on other data — here only the **x-coordinate** is selected ∇

Δ Primary selection asks what demands this data?

non_renewables										
	coalCap	coalOut	country	gasCap	gasOut	nuclearCap	nuclearOut	petrolCap	petrolOut	year
➡	405.5	345.13	JPN	747.84	425.37	333.23	49.11	241.69	53.78	2018

renewables					
▶ capacity country energyType output year					
225	18.22	JPN	Bio	30.13	2018
226	246.51	JPN	Hydro	81.11	2018
227	491.96	JPN	Solar	62.11	2018
228	30.66	JPN	Wind	6.44	2018

non_renewables										
	coalCap	coalOut	country	gasCap	gasOut	nuclearCap	nuclearOut	petrolCap	petrolOut	year
57	405.5	345.13	JPN	747.84	425.37	333.23	49.11	241.69	53.78	2018

renewables					
▶ capacity country energyType output year					
225	18.22	JPN	Bio	30.13	2018
226	246.51	JPN	Hydro	81.11	2018
227	491.96	JPN	Solar	62.11	2018
228	30.66	JPN	Wind	6.44	2018

Adding **capacity** for any renewable makes **nuclearGen** and all renewable **output** relevant too — because these are demanded by the **y-coordinate**


```

let nonRenewables x = x.nuclearCap + x.petrolCap + x.gasCap + x.coalCap;
  isCountry name x = name == x.country;
  isYear year x = year == x.year;

let energy year c_names =
  let rens = filter (isYear year) renewables;
    nonrens = filter (isYear year) non_renewables;
  let energy_per_country c_name =
    let rens' = filter (isCountry c_name) rens;
      non_ren_x = head (filter (isCountry c_name) nonrens);
      rensOut = sum (map (fun x → x.output) rens');
      rensCap = sum (map (fun x → x.capacity) rens');
      nonRensCap = nonRenewables non_ren_x
    in {
      name: c_name,
      totalCap: rensCap + nonRensCap,
      non_renewables: nonRensCap,
      renCapFactor:
        (rensOut + non_ren_x.nuclearOut) / (rensCap + non_ren_x.nuclearCap)
    }
  in map energy_per_country c_names;
let countries = [
  "BRA", "CHN", "DEU", "FRA", "EGY", "IND", "JPN", "MEX", "NGA", "USA"
]
in ScatterPlot {
  caption: "Clean energy efficiency vs. proportion of renewable energy cap",
  data: [{
    x: ctry.non_renewables / ctry.totalCap,
    y: ctry.renCapFactor, c: ctry.name
  } | ctry ← energy 2018 countries
],
  xlabel: "Renewables/TotalEnergyCap",
  ylabel: "Clean Capacity Factor"
}

```

Fluid programming language

Untyped, interpreted, functional

Records, dictionaries, data types, matrices

Programs evaluate to result plus

dynamic dependence graph G

Reachability in G and G^{-1} provides demands
and demandedBy operators ∇ and Δ

d3.js front end with “data linking” built-in

Only “data-transparency” so far

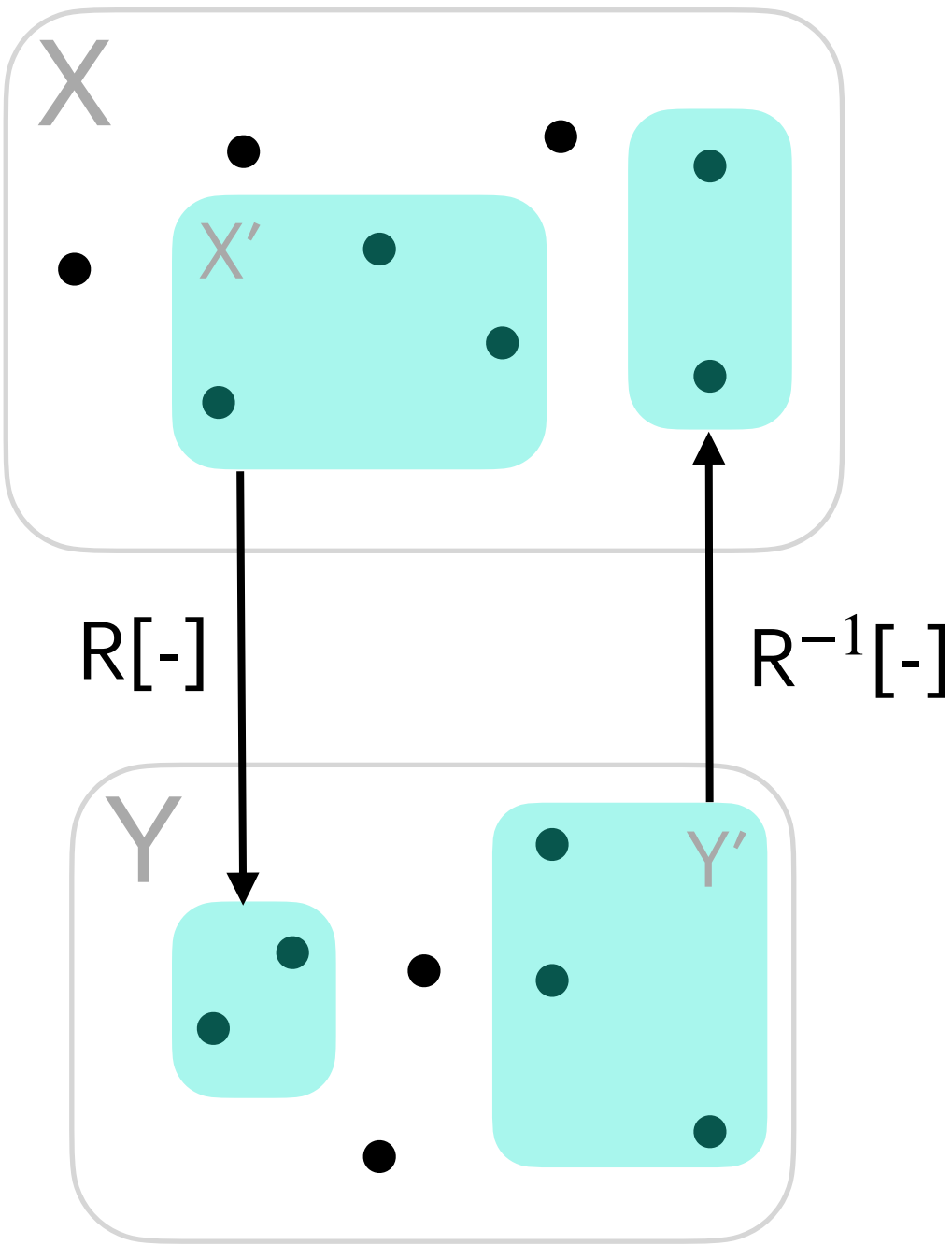
Intensional transparency is future work!

Conjugate operators for relations of “cognacy”

Suppose $R \subseteq X \times Y$

Consider **image** and **preimage** functions between powersets

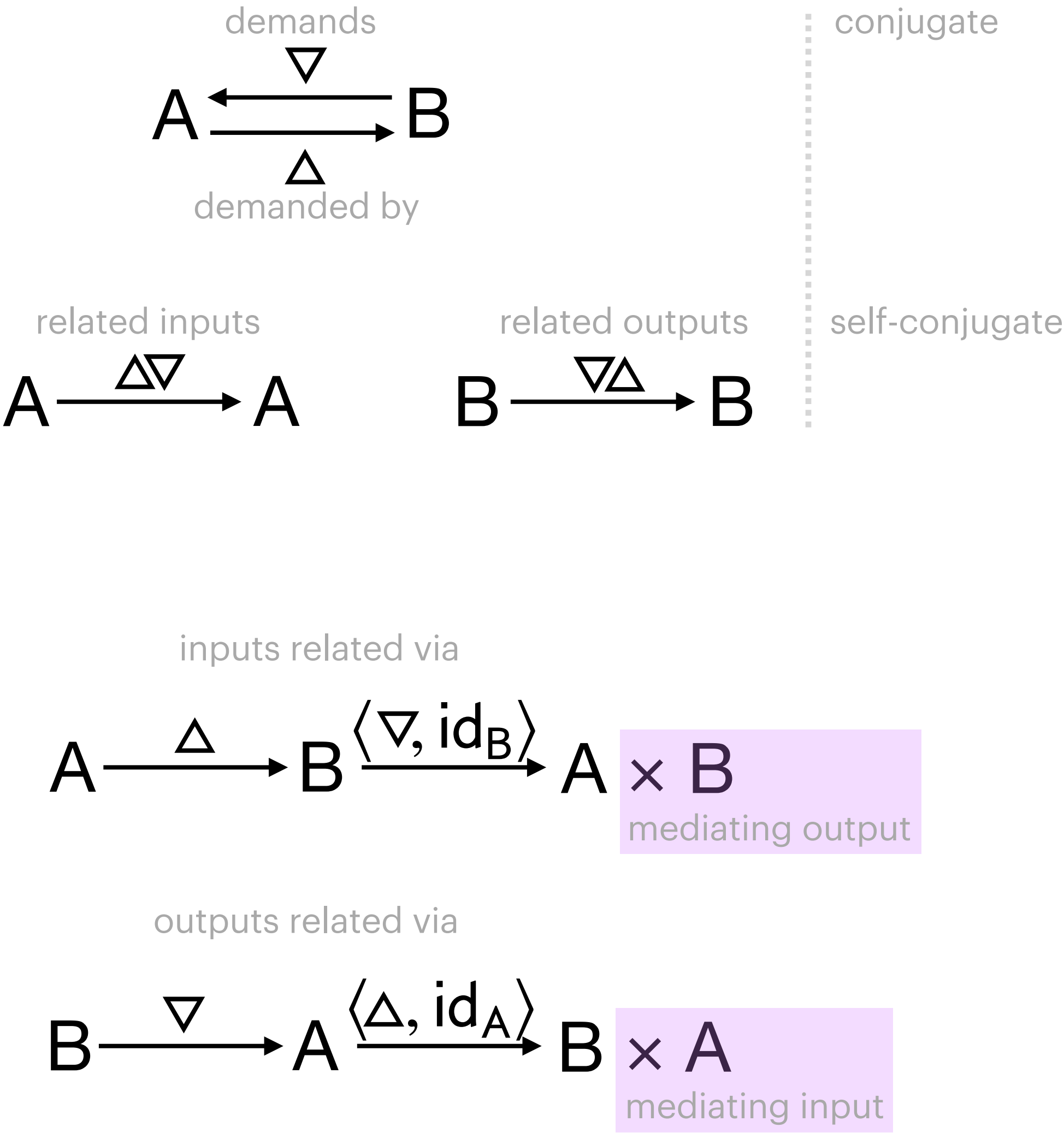
$$\mathcal{P}X \begin{matrix} \xleftarrow{R[-]} \\ \xrightarrow{R^{-1}[-]} \end{matrix} \mathcal{P}Y$$



These are **conjugate**:

$$R[X'] \cap Y' = \emptyset \quad \text{iff} \quad X' \cap R^{-1}[Y'] = \emptyset$$

In Fluid R is **reachability** (∇) in the dependence graph



Related outputs, revisited

Related inputs problem is dual to **brushing and linking** in data viz

Explored in POPL 2022

Swap one operator by its conjugate to compute related outputs:

$$C \xrightarrow{\nabla_f} A \xleftarrow{\nabla_g} B$$

$$C \xleftarrow{\Delta_f} A \xleftarrow{\nabla_g} B$$

But doesn't lend itself to related inputs:

$$A \xleftarrow{\nabla_f} C \text{ ⚡ } B \xleftarrow{\Delta_g} A$$

More general form is composition of **product view** $\langle f, g \rangle$ with its own conjugate

$$B \times C \xleftarrow{\Delta} A \xleftarrow{\nabla} B \times C$$

with (optional) **projections** to disregard irrelevant views

$$C \xleftarrow{\Delta_{\text{snd}}} B \times C \xleftarrow{\Delta} A \xleftarrow{\nabla} B \times C \xleftarrow{\nabla_{\text{fst}}} B$$

Projection clamps **demand** of discarded part at $\perp$

$$\nabla_{\text{fst}}(b) = (b, \perp) \qquad \Delta_{\text{fst}}(b, c) = b$$

Recovers related outputs but amenable to related inputs

$$A \xleftarrow{\nabla} B \times C \xleftarrow{\Delta} A$$

Related outputs with focusing

Projection operators turn out to be important for focusing queries around criteria of interest

Unfocused query

output

5	4	2	5	2
3	1	2	-1	-2
3	0	1	0	-1
2	1	-2	0	0
1	0	-1	-1	-2

filter

-2	-1	0
-1	1	1
0	1	2

input_image

15	13	6	9	16
12	5	15	4	13
14	9	20	8	1
4	10	3	7	19
3	11	15	2	9

Related outputs with respect to both inputs

Focused query

output												
5	4	2	5	2								
3	1	2	-1	-2								
3	0	1	0	-1								
2	1	-2	0	0								
1	0	-1	-1	-2								
					15	13	6	9	16			
					12	5	15	4	13			
					14	9	20	8	1			
					4	10	3	7	19			
					3	11	15	2	9			

Related outputs with respect to input_image only

Compose ∇ with operator which discards filter
Clamps availability of filter at $\top$

Related inputs with focusing

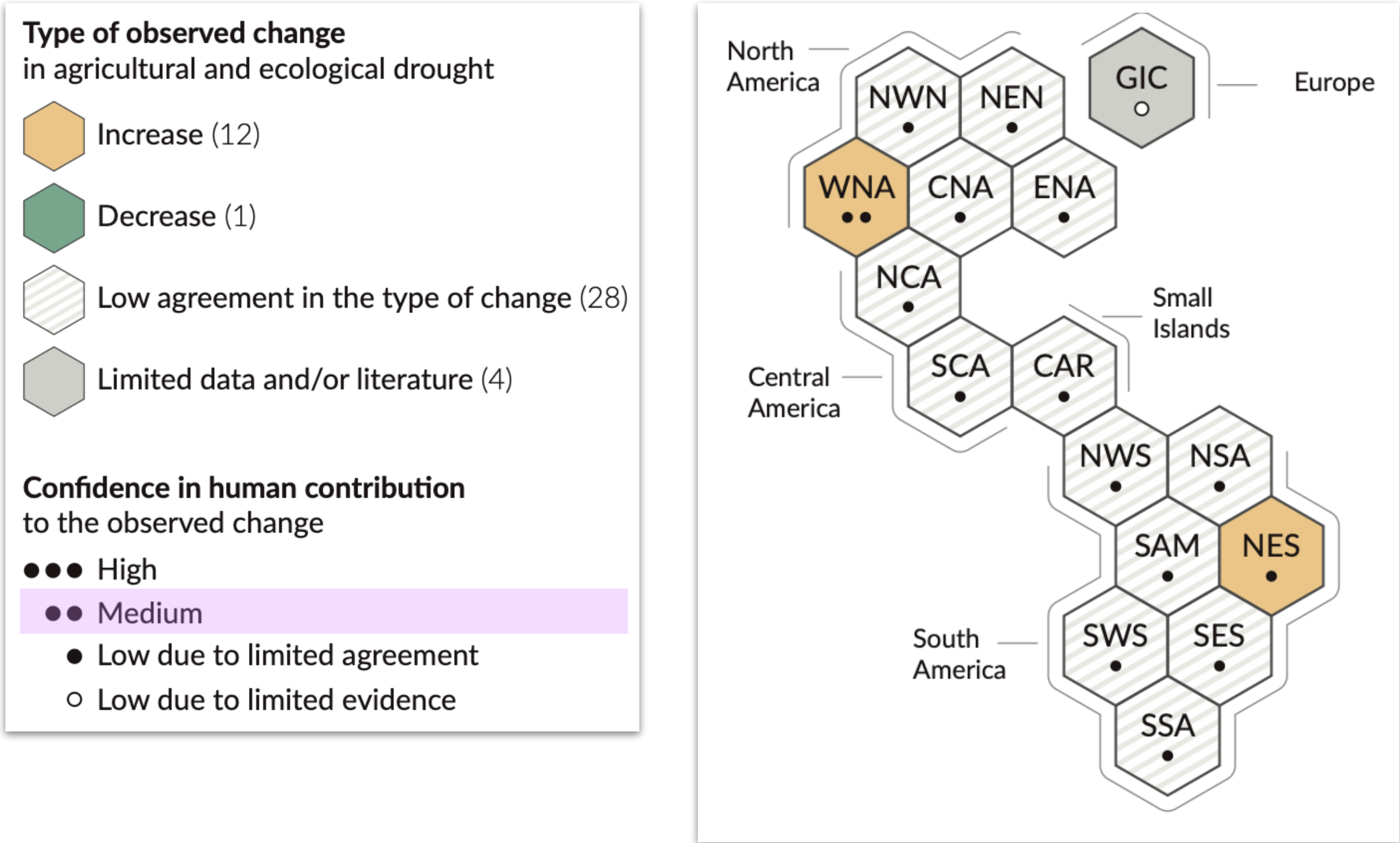
Compose ∇ with operator that discards irrelevant output

For example:

x or y coordinate of single point
layer of visualisation
entire chart from 1 or more charts

Clamps demand at $\perp$

Illustrative use case



Here we might focus on confidence in human contribution to fix query context for exploring relevant inputs and relationships between inputs

What next

Language

Markdown + quasiquotation
Units of measure
Multidimensional arrays
Type system?
Effect system?

Analysis

Intensional explanations
Better dependency relation

<http://f.luid.org>

<https://github.com/explorables-viz/fluid>

Applications

Online articles and long-form essays
presenting environmental science with
transparent and self-explanatory charts and
figures



Institute of
Computing for
Climate Science